

Sagan Chowdhury

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EXPERIENCE

AI Research Assistant

March 2026 – Present

California State University - East Bay

Hayward, CA

- Designing and evaluating prompt engineering frameworks for safe, reliable LLM deployment in production environments
- Investigating adversarial attack surfaces (prompt injection, jailbreaks) and mitigation strategies for GenAI systems; analyzing evaluation results to catch regressions across framework iterations

Software & Systems Intern

May 2026 – August 2026

Women's Cancer Resource Center

Berkeley, CA

- Designing and implementing automated, event-driven workflows with Microsoft Power Automate and Microsoft 365, orchestrating multi-step processes that streamline internal operations org-wide
- Building digital forms and end-to-end process pipelines with Adobe Acrobat Sign and SharePoint, reducing manual overhead and error rates across org-wide programs
- Owning the technical backlog for internal services: triaging and debugging issues, proposing solutions, and shipping systems improvements with measurable impact on staff efficiency

Teaching Assistant – Intro to Python

January 2024 – May 2024

City College of San Francisco

San Francisco, CA

- Supported 30+ students per semester with live debugging, code review, and concept reinforcement in Python during lab sections

PROJECTS

Reliflow – Reliability-First Agent Orchestration Framework | *Python, Pydantic v2, LiteLLM, SQLite, Typer* | [GitHub](#)

- Designing and shipping a pip-installable Python framework for orchestrating multi-step LLM agent workflows, with mandatory reflexion, LLM-verified plans, and hard cost guards enforced by the execution model
- Architected an event-sourced state store over SQLite with time-travel replay and a structured JSONL trace emitter/reader – observability and reproducible debugging of failed runs baked into the runtime
- Enforced correctness with a strictly layered package (no upward imports), pytest suite using injectable fake LLM clients (no network in CI), and ruff + mypy gating; shipped a Typer CLI (`init/run/trace`) delivering install-to-completed-run in under 15 minutes

Candor – AI-Powered News Bias Detection | *TypeScript, Next.js, Anthropic API, Supabase, Zod* | [GitHub](#)

- Built and deployed a full-stack Next.js web service performing real-time rhetorical and bias analysis of news articles via the Anthropic Messages API, achieving sub-10-second end-to-end response times
- Engineered a serverless, distributed content-extraction pipeline (Mozilla Readability + JSDOM on Vercel edge), enabling analysis of arbitrary URLs with no dedicated backend
- Enforced deterministic, schema-validated AI output via structured prompting, Zod validation, and retry logic on parse failure – improving reliability and eliminating hallucinated responses

Local LLM Desktop Client | *TypeScript, React, Electron, Ollama* | [GitHub](#)

- Built a cross-platform Electron/React desktop client running LLMs fully offline via Ollama, with a real-time streaming chat interface and multimodal image-attachment support across multiple local model backends
- Packaged and distributed standalone binaries for macOS and Windows with auto-detection of the Ollama runtime

EDUCATION

California State University East Bay

Hayward, CA

Bachelor of Science in Computer Science – GPA: 4.0

Aug. 2025 – May 2027

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL | **Concepts:** distributed systems, workflow orchestration, serverless, adversarial testing

Web Technologies: Next.js, React, Flask, FastAPI, Django, Electron, REST APIs, Vercel edge functions

Infrastructure & Tools: Linux/Unix CLI, Docker, Git/GitHub, Typer/CLI tooling, workflow automation (Power Automate, n8n, Make.com)

Databases & Datastores: PostgreSQL, SQLite, Supabase (relational); JSON

Data & AI/ML: Pandas, NumPy, scikit-learn, NLTK, TF-IDF, Zod, JSDOM, Anthropic / OpenAI / Gemini APIs